

Yijie Xu 徐亦捷

First Year Ph.D. Candidate in Artificial Intelligence at
The Hong Kong University of Science and Technology (Guangzhou)
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RESEARCH INTEREST

His current research interest is **Trustworthy Large Language Models**, he also has broad research interests focusing on Human-AI Interaction and Computational Social Science.

EDUCATION

Hong Kong University of Science and Technology (Guangzhou) *Guangzhou, China*
PH.D. STUDENT IN ARTIFICIAL INTELLIGENCE *Aug. 2027 Expected*

- **Supervisor:** Prof. Hui Xiong, Associate Vice President, Founding Head of AI Thrust, Fellow of AAAS, AAAI, IEEE.
- **Co-supervisor:** Prof. Xuming Hu, Assistant Professor in AI Thrust, HKUST(GZ).

Hong Kong University of Science and Technology (Guangzhou) *Guangzhou, China*
M.PHIL. IN ARTIFICIAL INTELLIGENCE *Sep. 2022 - Aug. 2024*

- **CGPA:** 3.96/4.3
- **Supervisor:** Prof. Hui Xiong, Associate Vice President, Founding Head of AI Thrust, Fellow of AAAS, AAAI, IEEE.
- **Co-supervisor:** Prof. Xiaojuan Ma, Associate Professor in CSE, HKUST.

University of Reading *Reading, England*
B.SC. IN MATHEMATICS AND APPLIED MATHEMATICS *Sep. 2018 - Jun. 2022*

- **CGPA:** 3.71/4
- **Final Year Project:** Research on Interpolation and Data Fitting: Basis and Applications.

Nanjing University of Information Science and Technology *Nanjing, China*
B.SC. IN MATHEMATICS AND APPLIED MATHEMATICS *Sep. 2018 - Jun. 2022*

- **Double First-class** University, also known as NUIST.
- **Dual-degree** with Department of Mathematics & Statistics, University of Reading.

SCHOLARSHIPS AND AWARDS

Doctoral Scholarship Award *Sep. 2024*
Postgraduate Scholarship Award *Sep. 2022*
First Prize of the *Smart Audiovisual Track*, National AI Application Innovation Challenge *Sep. 2024*
Champion of the *HCI channel*, the 2nd Guangzhou Pazhou Algorithm Competition (**Top 15/3307**) *Sep. 2023*
Scholarship from Nanjing University of Information Science and Technology *Nov. 2021*
Scholarship from China Mobile Communications Corporation (CMCC) *Nov. 2020*
Scholarship from China Mobile Communications Corporation (CMCC) *Nov. 2019*

TEACHING EXPERIENCES

Spring 2025 **Big Data Applications for Business and Government**, Teaching Assistant *HKUST(GZ)*
Fall 2024 **Natural Language Processing and Its Applications**, Teaching Assistant *HKUST(GZ)*
Summer 2023 **Red Bird Challenge Camp**, Teaching Assistant *HKUST(GZ)*
Summer 2022 **Red Bird Challenge Camp**, Teaching Assistant *HKUST(GZ)*
Fall 2021 **Professional Skills for Mathematicians**, Teaching Assistant *NUIST*
Fall 2021 **Numerical Analysis**, Teaching Assistant *NUIST*
Fall 2021 **Applied Stochastic Process**, Teaching Assistant *NUIST*
Spring 2021 **Mathematical Programming**, Teaching Assistant *NUIST*

PUBLICATIONS

†: Co-first Authors; *: Corresponding Authors.

CONFERENCES

- [1] **Mark Your LLM: Detecting the Misuse of Open-Source Large Language Models via Watermarking**
Yijie Xu[†], Aiwei Liu[†], Xuming Hu*, Lijie Wen, Hui Xiong*
WMARK@ICLR2025: *The 1st Workshop on GenAI Watermarking, ICLR 2025*
- [2] **SePer: Measure Retrieval Utility Through The Lens Of Semantic Perplexity Reduction**
Lu Dai, Yijie Xu, Jinhui Ye, Hao Liu*, Hui Xiong*
ICLR'25 Spotlight(3.20%): *The Thirteenth International Conference on Learning Representations*
- [3] **SpGesture: Source-Free Domain-adaptive sEMG-based Gesture Recognition with Jaccard Attentive Spiking Neural Network**
Weiyu Guo, Ying Sun*, Yijie Xu, Ziyue Qiao, Yongkui Yang, Hui Xiong*
NeurIPS'24: *The Thirty-eighth Annual Conference on Neural Information Processing Systems*
- [4] **Revisiting Noise Resilience Strategies in Gesture Recognition: Short-Term Enhancement in sEMG Analysis**
Weiyu Guo, Ziyue Qiao, Ying Sun*, Yijie Xu, Hui Xiong*
ICML'25: *The Forty-Second International Conference on Machine Learning*
- [5] **Event Camera Demosaicing via Swin Transformer and Pixel-focus Loss**
Yunfan Lu, Yijie Xu, Wenzong Ma, Weiyu Guo, Hui Xiong*
CVPRW'24: *Workshop for IEEE Conference on Computer Vision and Pattern Recognition*
- [6] **MIPI 2024 Challenge on Demosaic for HybridEVS Camera: Methods and Results**
Yaqi Wu, *et al.*, Yunfan Lu, Yijie Xu, Wenzong Ma, Weiyu Guo, Hui Xiong, *et al.*
CVPRW'24: *Workshop for IEEE Conference on Computer Vision and Pattern Recognition*
- [7] **Logic-in-Frames: Dynamic Keyframe Search via Visual Semantic-Logical Verification for Long Video Understanding**
Weiyu Guo, Ziyang Chen, Shaoguang Wang, Jianxiang He, Yijie Xu, Jinhui Ye, Ying Sun*, Hui Xiong*
NeurIPS'25: *The Thirty-ninth Annual Conference on Neural Information Processing Systems*
- [8] **See&Trek: Training-Free Spatial Prompting for Multimodal Large Language Model**
Pengteng Li, Pinhao Song, Wuyang Li, Huizai Yao, Weiyu Guo, Yijie Xu, Dugang Liu, Hui Xiong
NeurIPS'25: *The Thirty-ninth Annual Conference on Neural Information Processing Systems*

JOURNALS

- [9] **Against Political Polarization: A Unified Framework for Tracing Evolving Political Ideologies on Social Media**
Yijie Xu, Chao Wang, Hui Xiong*
TIST: *ACM Transactions on Intelligent Systems and Technology*. (Under 2nd round revision)

PREPRINTS

- [10] **You only need 4 extra tokens: Synergistic Test-time Adaptation for LLMs**
Yijie Xu, Huizai Yao, Zhiyu Guo, Weiyu Guo, Pengteng Li, Aiwei Liu, Xuming Hu, Hui Xiong
Under Review for ICLR'26
- [11] **VSI: Visual-Subtitle Integration for Keyframe Selection to Enhance Long Video Understanding**
Jianxiang He, Shaoguang Wang, Weiyu Guo, Meisheng Hong, Jungang Li, Yijie Xu, Ziyang Chen, Hui Xiong
arXiv
- [12] **Less is More: Token-Efficient Video-QA via Adaptive Frame-Pruning and Semantic Graph Integration**
Shaoguang Wang, Jianxiang He, Weiyu Guo, Ziyang Chen, Yijie Xu, Hui Xiong
arXiv

- [13] **Beyond Boundaries: Leveraging Vision Foundation Models for Source-Free Object Detection**
Huizai Yao, Sicheng Zhao, Pengteng Li, Yi Cui, Shuo Lu, Weiyu Guo, Yunfan Lu, Yijie Xu, Hui Xiong
Under Review
- [14] **Contrastive Graph Matching for Domain Adaptive Object Detection**
Pengteng Li, Huizai Yao, Yi Cui, Yunfan Lu, Yijie Xu, Minghao Yang, Pinhao Song
arXiv

SERVICES

- **Reviewer**
 - Conferences:
 - * **2025**: AAAI, ACM MM, KDD, NeurIPS, ICML, IJCAI, ICLR, ICASSP, AISTATS, ICWSM
 - * **2024**: NeurIPS
 - Journals: IEEE Transactions on Circuits and Systems for Video Technology
- **Other Roles**
 - Student Representative Member, Committee on Undergraduate Studies, Senate of HKUST(GZ)

SKILLS

- **Language**: Chinese (Mother Tongue), English (IELTS:7)
- **Hobbies**: Photography, Badminton, Swimming, Piano